

A Profile-Parameterized Stopping Target

10D BO now *strictly dominates* the 6D champion: $+8\% S/\sqrt{B}$ at -12% flash

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Mu2e — autoresearch / closed-loop Bayesian optimization

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Result in one slide

Free the **whole** stopping target — quadratic profiles for radius, thickness, and hole over 49 foils, plus total extent — and BO finds a design that beats the 6D champion on **both** objectives at once, on corrected geometry.

- **80 evaluations**, four campaigns (foilspf01–04), every row cleared a strict **zero-volume-overlap** preflight gate.
- foilspf03R02_00: $S/\sqrt{B}$ **4.41** vs the re-measured 6D champion's 4.10, at flash 9.2×10^{-7} vs 1.04×10^{-6} — **no trade-off**, it wins on both axes.
- A second point, foilspf04R03_00, holds champion $S/\sqrt{B}$ (4.07) at the **deployed target's own flash** — +25% significance for free.
- The flash frontier now spans **25×**: floor 5.9×10^{-8} MeV/POT, $\sim 10\times$ below anything the 6D line reached.

All numbers are **Run1Bap-scale**: a deliberate 2026-05 filter-acceptance recovery upstream lifts $S/\sqrt{B}$ +5% vs older (Run1Bak) results — the old 3.90 ceiling reads ≈ 4.10 here.

What we optimize

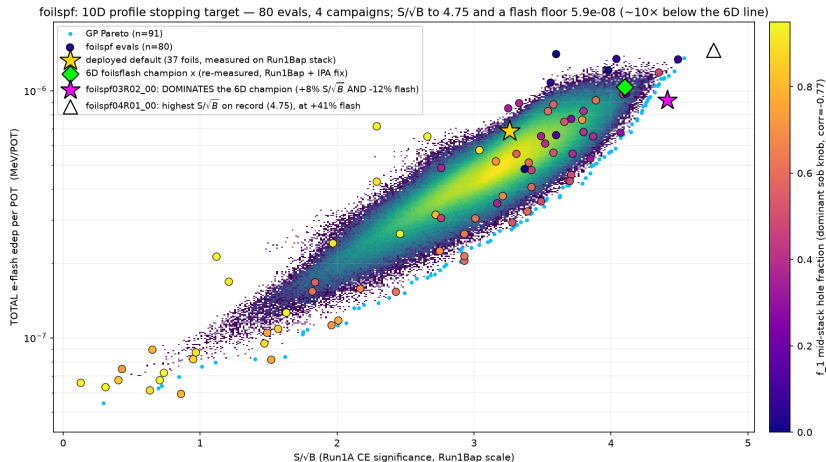
10-D profile parameterization: the 49-foil stack is generated from $K=3$ quadratic control points for each of outer radius, half-thickness, and hole fraction, plus a total-extent knob — no fixed base, geometrically subsumes the deployed target.

knob	range	
$r_{\text{Out-}\{0,1,2\}}$	50–120	mm
$h_{\text{T-}\{0,1,2\}}$	0.01–0.15	mm
$f_{\{0,1,2\}}$	0–0.95	hole fraction
extent	400– 2000	mm (deployed: 800)

objective	direction
$S/\sqrt{B}$ (Run1A CE)	maximize
e^- flash edep per POT in tracker	minimize

Same closed loop and objectives as the 6D (foilsflash) line; **corrected stack:** Run1Bap release, inner-proton-absorber placement fix, and a strict *zero-overlap* geometry gate every 6D row predates. The extent ceiling was raised 1100 $\rightarrow$ 1700 $\rightarrow$ 2000 mid-programme (next slide but one).

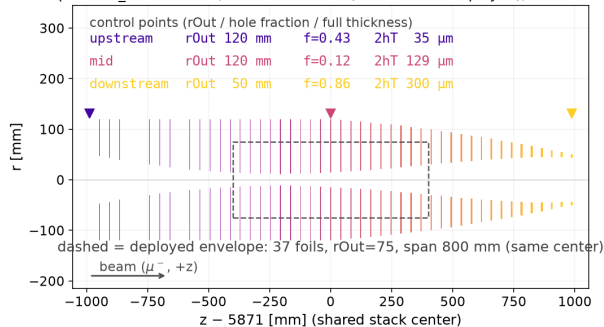
The landscape



80 evaluations, GP cloud over the 10D box. Gold star = deployed default; green diamond = 6D champion geometry re-measured on this stack; **magenta star** = foilsfpf03R02_00, below and to the right of the diamond — strictly better. Colour = mid-stack hole fraction, still the dominant $S/\sqrt{B}$ knob ($\text{corr} -0.77$ at $n=80$): open mid holes shed stopping

Anatomy of the best point

foilsfp03R02_00 — 49 foils, extent 1975 mm (± 588 mm vs deployed), thickness $\times 30$



design	$S/\sqrt{B}$	flash
03R02_00 (champion)	4.41	0.92
04R01_00 (max $S/\sqrt{B}$)	4.75	1.46
04R03_00 (flash-matched)	4.07	0.68
6D champion $\times$ (re-meas.)	4.10	1.04
deployed default	3.26	0.69
02R02_00 (flash floor)	0.86	0.059

flash in 10^{-6} MeV/POT, all measured on the Run1Bap stack

- Wide, nearly-solid discs (rOut 120 mm, $f \rightarrow 0.12$) through the upstream half; tapering to **small thick annuli** downstream (rOut 50 mm, $f=0.86$, 300 μm).
- Extent 1975 mm — ± 588 mm past the deployed span, growth **symmetric** about the shared center ($z_0=5871$), dashed = deployed envelope.
- Shape flipped vs the earlier 1015 mm champion, which put its mass *downstream*. Mass-forward wins once the stack is long.

The extent knob — and what we do *not* know

What raising the ceiling bought

- foilspf02 pinned the old 1100 mm bound: its two best new front points both sat **exactly on it**.
- Ceiling re-measured by live worst-corner preflight: 1400, 1700 pass; 1800 fails. Moving one **massless** bookkeeping volume (EMC_Source, a 20 μm vacuum disc no stage reads) reopened 1900 and 2000; 2100 fails.
- Rule adopted: extend only by relocating **massless** volumes, **never** real material.
- Split of the $S/\sqrt{B}$ gain: +0.32 from the higher bound, +0.34 from more search. Roughly half each.

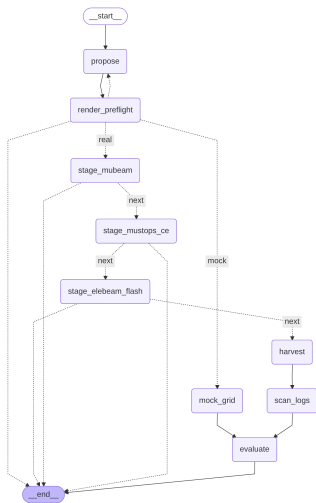
Why longer helps

- Extent is **orthogonal to material by construction**: each foil's radius, thickness and hole are functions of its *index* $i/48$, never of z . Extent sets only the pitch $\Delta z = \text{extent}/48$ (8.3 mm at 400, 41.7 mm at 2000; deployed 22.2 mm) — a longer stack is the *same foils spread further apart* (champion: 183 cm^3 at any length).
- The gain is therefore in **stopping** (corr +0.35); acceptance *per stop* actually falls (corr -0.37). A longer stack intercepts more of the muon beam's range spread.

Open — no turnover observed

15 of 80 rows sit at *exactly* 2000: the optimizer is still pushing at the wall. 2000 is where the **zero-overlap corridor** ends, not where the physics turns over — **no proof it is optimal**. A fixed-shape scan would settle it.

Machinery (unchanged)



Same closed loop as the 6D line — generated from the running code (`draw_mermaid()`, `foilspf` mode):

- One LangGraph run = one evaluation: `propose` → `preflight` → 3 grid stages → `harvest` → `row`.
- hybrid picker (qNEHVI + qNParEGO), $q=10$ in flight, rolling to 20 evals per campaign; GP refits as rows land.
- Preflight **requires zero overlaps** — reachable only on this corrected stack.
- No row is ever written from partial results: 80 launched, 80 landed.

```
python -m graph.closed.loop --mode foilspf \  
--picker hybrid --q 10 --rolling --max-evals 20
```

Conclusion & next steps

- The profile parameterization is **strictly stronger** than the 6D extras — not merely competitive: it wins on $S/\sqrt{B}$ and flash simultaneously, and reaches a flash floor $\sim 10\times$ lower.
- The most deployable point may not be the champion: foilspf04R03_00 offers +25% $S/\sqrt{B}$ **over the deployed target at the deployed target's own flash.**
- Still on **clean geometry**: correct absorber position, zero overlaps — no 6D row can claim either.
- **Running now**: foilspf2k01 — 9D *shape-only* at extent pinned to 2000, to ask how much is left in shape once length is fixed.
- **Needed**: a controlled extent scan at fixed shape (400...2400) to find the actual optimum — the geometry corridor reaches ~ 2400 if the stack is also shifted upstream. And a 400-job flash confirmation of foilspf03R02_00.